

Table of Trigonometric Identities and Basic Functions

Trigonometric Identities	
$e^{\pm j\theta} = \cos(\theta) \pm j \sin(\theta)$	$\sin(\alpha \pm \beta) = \sin(\alpha) \cos(\beta) \pm \cos(\alpha) \sin(\beta)$
$\cos(\theta) = 0.5[e^{j\theta} + e^{-j\theta}]$	$\cos(\alpha \pm \beta) = \cos(\alpha) \cos(\beta) \mp \sin(\alpha) \sin(\beta)$
$\sin(\theta) = -0.5j[e^{j\theta} - e^{-j\theta}]$	$\tan(\alpha \pm \beta) = \frac{\tan(\alpha) \pm \tan(\beta)}{1 \mp \tan(\alpha) \tan(\beta)}$
$\sin^2(\theta) + \cos^2(\theta) = 1$	
$\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$	$\sin(\alpha) \sin(\beta) = 0.5[\cos(\alpha - \beta) - \cos(\alpha + \beta)]$
$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$	$\cos(\alpha) \cos(\beta) = 0.5[\cos(\alpha - \beta) + \cos(\alpha + \beta)]$
$\sin^2(\theta) = 0.5[1 - \cos(2\theta)]$	$\sin(\alpha) \cos(\beta) = 0.5[\sin(\alpha - \beta) + \sin(\alpha + \beta)]$
$\cos^2(\theta) = 0.5[1 + \cos(2\theta)]$	$C \cos(\theta) - S \sin(\theta) = (C^2 + S^2)^{0.5} \cos\left[\theta + \tan^{-1}\left(\frac{S}{C}\right)\right]$
Complex Unit: $\left[j = (-1)^{0.5} = e^{j\pi/2} = e^{j90^\circ} \right] \quad \left[-j = j^{-1} = e^{-j\pi/2} = e^{-j90^\circ} \right]$ $\left[j^2 = -1 = e^{j\pm 180^\circ} \right]$	

Definitions of Basic Functions
Rectangle: $\text{rect}\left(\frac{t}{T}\right) = \begin{cases} 1; & -T/2 \leq t < T/2 \\ 0; & \text{elsewhere} \end{cases}$
Triangle: $\text{tri}\left(\frac{t}{T}\right) = \begin{cases} 1 - t /T; & t \leq T \\ 0; & t > T \end{cases}$
Sine Cardinal: $\text{sinc}\left(\frac{t}{T}\right) = \begin{cases} \frac{\sin(\pi t/T)}{\pi t/T}; & t \neq 0 \\ 1; & t = 0 \end{cases}$
Signum: $\text{sgn}(t) = \begin{cases} 1; & t \geq 0 \\ -1; & t < 0 \end{cases}$
Unit Impulse: $\delta(t) = \begin{cases} \infty; & t = 0 \\ 0; & t \neq 0 \end{cases} \quad \int_{0^-}^{0^+} \delta(t) dt = 1$
Unit Step: $u(t) = \begin{cases} 1; & t \geq 0 \\ 0; & t < 0 \end{cases}$